Classical capacities and complementary simulation for qubit Davies channels

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Abstract

We extend the thermal block entropy estimate for generalized amplitude damping to the full qubit Davies region, including additional dephasing. The proof combines the triangular inequality of Tang, Zhu, Bai and Wang with a controlled-purification consequence of strong subadditivity: a third vertex completes the fixed-entropy decomposition used for the thermal boundary. The resulting population-resolved bound holds with every finite-dimensional partner. It implies strong superadditivity of the convex roof of output entropy, and hence arbitrary-partner additivity of Holevo information and classical capacity for both the Davies channel and its complementary channel. We give scalar convex-envelope formulas for both capacities. Every bipartite state supported on the two-dimensional Stinespring range has strongly additive entanglement of formation, so its state entanglement cost requires no regularization. A scalar moment program evaluates that formation functional for arbitrary input coherence. The state/channel correspondence, complement duality and existing axial convex-roof geometry are explicitly attributed to prior work. A normalized reverse channel stays within the Davies region, allowing us also to remove regularization from the parallel entanglement cost of the complementary channel. We obtain an elementary rate in a known flat-roof region and exhibit a nonunital example whose simulation cost strictly exceeds the cost of its Choi state. Finite-block and numerical bounds support the results. The triangular theorem remains an external dependency; the original channel's simulation cost, adaptive simulation and quantum capacity are not determined.

1 Channel family, antecedents and result

A qubit can relax toward a thermal population while its coherence decays faster than relaxation alone requires. Write the corresponding channel as

$$\Phi_{a,b,\lambda}\begin{pmatrix} 1-p & z \\ \bar{z} & p \end{pmatrix} = \begin{pmatrix} 1-t(p) & \lambda z \\ \lambda\bar{z} & t(p) \end{pmatrix}, \qquad t(p) = a + kp, \quad k = 1-a-b, \tag{1}$$

where

$$a, b \geq 0, \qquad a+b \leq 1, \qquad 0 \leq \lambda^2 \leq k, \qquad \lambda \geq 0. \tag{2}$$

A fixed phase of the coherence multiplier can be removed by an output unitary. The map is completely positive: it is generalized amplitude damping followed by dephasing with multiplier $\lambda/\sqrt{k}$ when $k > 0$. The limiting case $k = 0$ is a constant diagonal-output channel.

The region (2) is the closure of the usual dissipative qubit Davies family, allowing degenerate limits and exchange of the energy labels. Roga, Fannes and Życzkowski characterized these maps and their single-use minimum output entropy [1]. At finite positive relaxation times their parameters satisfy

$$k = e^{-t/T_1}, \qquad \lambda = e^{-t/T_2}, \qquad T_2 \leq 2T_1.$$

Here t in the exponents is time, whereas $t(p)$ in (1) is an output population. We use (1) as the mathematical definition throughout. General phase-covariant qubit channels allow the larger complete-positivity region $\lambda^2 \le k + ab$; that entire region is not claimed here.

The proof builds on two recent results. Tang et al. established strong additivity for qubit channels admitting a pure output, using a triangular block entropy inequality [2]. OpenAI extended that inequality to the thermal equality case and obtained strong additivity for generalized amplitude damping [3]. In our parameters that boundary is $\lambda^2 = k$. We add a third vertex to their fixed-entropy decomposition, proving the inequality for $\lambda^2 < k$ as well. The new vertex follows directly from strong subadditivity, and is not a new entropy axiom. The scalar convex envelope is necessary because extra dephasing need not preserve convexity of the pure-output entropy.

Several overlaps are already understood: unital qubit channels [5], entanglement-breaking channels [6], and the pure-output boundary [2]. Earlier work treated approximate Holevo additivity for damping combined with dephasing [7]; its ordinary damping examples now also fall within the pure-output theorem. Our contribution is the stated block extension and its additivity consequences across the thermal interior. Roga explicitly listed Davies capacity additivity as open in 2011 [11, Section 6, p. 78]; this historical statement is not a certificate of present priority. We do not claim a new description of Davies dynamics or a new single-qubit minimum-entropy optimization. The state formation and complementary-channel consequences below use the established Matsumoto–Shimono–Winter correspondence [12], complement duality [15], and the regularized formation characterization of state cost [13]. Hellmund and Uhlmann already analyzed the axial one-use convex roofs [16]; the new operational step is the stated strong additivity, not rediscovery of those roofs.

The simulation result concerns the complementary channel and the parallel diamond-norm task of Berta et al. [19]. Wilde already determined the cost of epolarizing channels, the complements of depolarizing channels, even under sequential tests [20, Section IV.D]. More generally, the unital qubit case is covered by the unital additivity and covariant channel/state machinery [5, 12, 20]. We do not claim these cases as new. The additional nonunital region is useful precisely because the optimizing reference input need not be a Bell pair; Section 6 gives an exact example. The scalar roof and the parallel simulation theorem are antecedents; closure of the explicit reverse parameters lets us apply the Davies range theorem to this different bipartition.

All spaces below are finite dimensional. Let $S(\rho) = -\operatorname{Tr}\rho\ln\rho$ be entropy in nats and put

$$h(p) = -p\ln p - (1-p)\ln(1-p), \qquad g(u) = h\left(\frac{1+\sqrt{1-4u}}{2}\right).$$

Zero terms are interpreted by continuity. Define

$$v(p) = t(p)(1-t(p)) - \lambda^2 p(1-p), \quad e(p) = g(v(p)), \quad f = \operatorname{co} e, \tag{3}$$

where $\operatorname{co} e$ is the largest convex function below e on $[0,1]$.

For a channel $\mathcal{N}$, $\chi(\mathcal{N})$ denotes the maximum ensemble Holevo information in bits, $S_{\min}(\mathcal{N}) = \min_\rho S(\mathcal{N}(\rho))$, and

$$C(\mathcal{N}) = \sup_{n\ge 1}\frac{1}{n}\chi(\mathcal{N}^{\otimes n})$$

is its unassisted memoryless classical capacity, with arbitrary entangled block encodings and collective output measurements [8, 9]. No preshared entanglement or feedback is assumed.

Theorem 1. *For every channel $\Phi = \Phi_{a,b,\lambda}$ satisfying (2),*

$$C(\Phi) = \chi(\Phi) = \frac{1}{\ln 2}\max_{0\le p\le 1}\{h(a+kp) - f(p)\}. \tag{4}$$

An ensemble with at most four pure signals, consisting of at most two equally weighted opposite-phase pairs within their respective populations, attains the one-use maximum. For every finite-dimensional channel $\mathcal{N}$,

$$\chi(\Phi \otimes \mathcal{N}) = \chi(\Phi) + \chi(\mathcal{N}), \tag{5}$$

$$S_{\min}(\Phi \otimes \mathcal{N}) = \min_p e(p) + S_{\min}(\mathcal{N}), \tag{6}$$

$$C(\Phi \otimes \mathcal{N}) = \chi(\Phi) + C(\mathcal{N}). \tag{7}$$

In particular $\chi(\Phi^{\otimes n}) = n\chi(\Phi)$ for every n.

Theorem 1 states unrestricted Holevo additivity. The stronger fixed-marginal convex-roof statement is proved in Section 4. The partner's classical capacity retains its regularization; we do not replace $C(\mathcal{N})$ by $\chi(\mathcal{N})$. Formula (4) is a scalar convex-envelope optimization, not a claimed elementary closed form.

2 The block entropy extension

We first state precisely the external inequality used in the proof.

Lemma 2 (Triangular entropy inequality, Tang et al.)**.** *Let A, B, C have the same finite rectangular shape, with squared Hilbert–Schmidt norms x, y, z summing to one. Then*

$$S\left(\begin{pmatrix} A & B \\ C & 0 \end{pmatrix}\begin{pmatrix} A & B \\ C & 0 \end{pmatrix}^*\right) \geq xS(AA^*/x) + yS(BB^*/y) + zS(CC^*/z) + g(yz).$$

A zero-weight term is omitted.

This is the same-shape rectangular case of [2, Theorem 4.2], after permutation of block rows, also stated and proved in [3, Theorem 2.1]. We invoke that result; it is not proved anew or attributed to this paper.

The additional vertex uses only data processing for conditional entropy. Its two-label case also follows from Amosov's phase-damping estimate [10, Proposition 5], with partial trace as the partner channel. We give the short purification proof to fix the direction of the entropy inequality; the vertex estimate itself is not claimed as new.

Lemma 3 (Controlled purifications)**.** *Let $\rho = (\rho_{ij})$ be a density matrix and let X_i be matrices of the same shape with $\operatorname{Tr} X_i X_i^* = 1$. Then*

$$S\left(\sum_{ij} \rho_{ij}|i\rangle\langle j| \otimes X_i X_j^*\right) \geq S(\rho) + \sum_i \rho_{ii} S(X_i X_i^*).$$

Proof. Vectorize X_i as a normalized vector $|X_i\rangle_{BE}$. The map $V|i\rangle = |i\rangle_A|X_i\rangle_{BE}$ is an isometry. Thus $\Omega = V\rho V^*$ has $S(ABE)_\Omega = S(\rho)$, and its AB marginal is the matrix in the statement. Dephase A, obtaining Ω'. Strong subadditivity implies that applying a channel to the conditioning system AB cannot decrease $S(E|AB)$ [4]. Consequently

$$S(AB)_\Omega - S(ABE)_\Omega \geq S(AB)_{\Omega'} - S(ABE)_{\Omega'}.$$

The right side is $\sum_i \rho_{ii} S(X_i X_i^*)$: the classical entropy of i cancels, and each conditional BE vector is pure. Rearrangement proves the claim. □

Theorem 4 (Thermal inequality with additional dephasing)**.** *Let* $X, Y \in \mathbb{C}^{d\times r}$ *satisfy* $\operatorname{Tr} XX^* = \operatorname{Tr} YY^* = 1$. *Let* $\alpha, \beta, \eta, \delta, K \geq 0$ *satisfy*

$$\alpha + \beta + \eta + \delta = 1, \qquad \alpha\delta - \beta\eta \geq K^2.$$

For

$$T = \begin{pmatrix} \alpha XX^* + \beta YY^* & KXY^* \\ KYX^* & \eta XX^* + \delta YY^* \end{pmatrix}, \quad U_0 = \alpha + \beta, \quad U_1 = \eta + \delta, \quad u = U_0U_1 - K^2,$$

one has

$$S(T) \geq (\alpha + \eta)S(XX^*) + (\beta + \delta)S(YY^*) + g(u). \tag{8}$$

Proof. The assumptions ensure positivity of T: its coherent part with diagonal blocks $\alpha XX^*, \delta YY^*$ is positive because $\alpha\delta \geq K^2$, and the remaining diagonal blocks are positive.

Assume first $U_0U_1 > 0$ and $u > 0$. With U_0, U_1, K fixed, the constraint becomes

$$\beta U_1 + \eta U_0 \leq u.$$

In the coefficient order $(\alpha, \beta, \eta, \delta)$ the feasible triangle has vertices

$$\begin{aligned} A &= (K^2/U_1, u/U_1, 0, U_1), \\ B &= (U_0, 0, u/U_0, K^2/U_0), \\ C &= (U_0, 0, 0, U_1). \end{aligned}$$

The original point has respective weights

$$w_A = \beta U_1/u, \qquad w_B = \eta U_0/u, \qquad w_C = 1 - w_A - w_B.$$

They are nonnegative. Each vertex retains the same row masses and off-diagonal scalar K, hence the same u.

At A, write

$$T_A = M_A M_A^*, \qquad M_A = \begin{pmatrix} \sqrt{\alpha}X & \sqrt{\beta}Y \\ \sqrt{\delta}Y & 0 \end{pmatrix}.$$

Here $\sqrt{\alpha\delta} = K$ and $\beta\delta = u$, so Lemma 2 gives (8) at this vertex. At B use

$$M_B = \begin{pmatrix} \sqrt{U_0}X & 0 \\ \sqrt{\delta}Y & \sqrt{\eta}X \end{pmatrix}.$$

It satisfies $M_B M_B^* = T_B$; interchange its block rows to apply the triangular lemma, with corner-mass product $U_0\eta = u$.

At C, apply Lemma 3 to $\rho = \left(\begin{smallmatrix} U_0 & K \\ K & U_1 \end{smallmatrix}\right)$ and the pair X, Y. The scalar state has determinant u and entropy $g(u)$, giving the desired bound. Finally, entropy concavity applied to $T = w_A T_A + w_B T_B + w_C T_C$ preserves $g(u)$ and the affine weights of the two block entropies.

If a row mass vanishes, $K = 0$ and the result is ordinary entropy concavity in the remaining block. If $u = 0$ and both row masses are positive, the constraint forces $\beta = \eta = 0$; Lemma 3 applies with a pure ρ. This also establishes all boundaries. □

The estimate is sharp for every allowed scalar coefficient tuple: setting $X = Y$ gives $T = \rho \otimes XX^*$, so equality holds. The equality boundary $\alpha\delta - \beta\eta = K^2$ is exactly the two-vertex thermal argument in [3, Theorem 3.1]; the third vertex is needed to cover the inequality region.

3 Population-resolved bound and capacity

Lemma 5. *Let $\mathcal{N}$ be any finite-dimensional channel and $|\psi\rangle = \sqrt{1-p}|0\rangle|\psi_0\rangle + \sqrt{p}|1\rangle|\psi_1\rangle$, where the nonzero branches are normalized. Then*

$$S((\Phi \otimes \mathcal{N})(\psi)) \geq e(p) + (1-p)S(\mathcal{N}(\psi_0)) + pS(\mathcal{N}(\psi_1)). \tag{9}$$

No orthogonality of the conditional vectors is required.

Proof. For Kraus operators L_j of $\mathcal{N}$, form the matrices with columns $X_j = L_j\psi_0$ and $Y_j = L_j\psi_1$. Their squared Hilbert–Schmidt norms are one. The output has the form in Theorem 4, with

$$\alpha = (1-p)(1-a), \quad \beta = pb, \quad \eta = (1-p)a, \quad \delta = p(1-b), \quad K = \lambda\sqrt{p(1-p)}.$$

Here $\alpha\delta - \beta\eta = kp(1-p) \geq K^2$, $\alpha + \eta = 1 - p$, and $u = v(p)$. Apply (8). At $p = 0, 1$ only one product branch remains and the formula follows directly. □

For clarity, the convex envelope can be written as

$$f(p) = \min_{\substack{0\leq q,r,s\leq 1 \\ (1-s)q+sr=p}} \{(1-s)e(q) + se(r)\}. \tag{10}$$

Indeed the minimum of $\int e\, d\mu$ over probability measures with mean p is attained by compactness. This minimum is the lower convex envelope. An extreme minimizing measure has at most two support points: weights on three distinct support points admit a nonzero signed perturbation preserving total mass and mean. Equivalently, each lower supporting line of the convex hull of the graph needs at most two contact points. Continuity of e includes endpoint and one-point cases.

Proof of Theorem 1. Refining any signal ensemble into pure states does not decrease its Holevo information. A pure qubit with population p has output entropy $e(p)$, independently of its phase. If an ensemble has mean population $\bar{p}$, dephasing its average output increases entropy, and convexity of $f \leq e$ gives

$$(\ln 2)\chi(\text{ensemble}) \leq h(a + k\bar{p}) - \sum_\ell w_\ell e(p_\ell) \leq h(a + k\bar{p}) - f(\bar{p}).$$

Conversely choose q, r, s attaining (10). At population q use $\sqrt{1-q}|0\rangle \pm \sqrt{q}|1\rangle$, each with probability $(1-s)/2$; at population r use the analogous pair, each with probability $s/2$. The average is diagonal with population p, and the average output entropy is $f(p)$. Maximizing over p proves the single-use formula and the four-signal bound.

For a pure ensemble $\{w_\ell, \psi_\ell\}$ on the input of $\Phi \otimes \mathcal{N}$, apply Lemma 5 to every signal. The branch outputs of $\mathcal{N}$ with probabilities $w_\ell(1-p_\ell), w_\ell p_\ell$ form a valid ensemble. Its average is exactly the second output marginal $\bar{\omega}_B$. Subadditivity bounds the entropy of the full average output by $h(a + k\bar{p}) + S(\bar{\omega}_B)$. Subtracting the average of (9) yields

$$(\ln 2)\chi(\text{joint ensemble}) \leq h(a + k\bar{p}) - f(\bar{p}) + (\ln 2)\chi(\mathcal{N}).$$

The single-use formula and product ensembles prove (5). The same one-step inequality bounds every pure output entropy below by $\min_p e(p) + S_{\min}(\mathcal{N})$; entropy concavity handles mixed inputs, and products attain the bound. This proves (6).

Iteration of (5) gives $\chi(\Phi^{\otimes n} \otimes \mathcal{N}^{\otimes n}) = n\chi(\Phi) + \chi(\mathcal{N}^{\otimes n})$. Taking the regularized capacity proves (7), including $C(\Phi) = \chi(\Phi)$. □

4 Entanglement formation in the dilation range

For any finite channel $\mathcal{M}$ define its convex roof of output entropy, in nats, by

$$\widehat{S}_{\mathcal{M}}(\rho) = \min_{\rho=\sum_j w_j\psi_j} \sum_j w_j S(\mathcal{M}(\psi_j)),$$

where the ψ_j are pure states. If $V : \mathbb{C}^2 \to B \otimes E$ is a Stinespring isometry of Φ, then

$$F_\Phi(\sigma) := \widehat{S}_\Phi(\sigma) = E_F(V\sigma V^*). \tag{11}$$

Here E_F denotes bipartite state entanglement of formation in nats. Equation (11) is the standard channel/state correspondence [12]. Each pure component of a state supported on $\operatorname{ran} V$ lies in that range and therefore pulls back to a pure qubit; this also gives a direct proof of the equality.

An elementary observation strengthens the population argument. Every qubit state $\sigma = \left(\begin{smallmatrix} 1-p & z \\ \bar{z} & p \end{smallmatrix}\right)$ is a mixture of pure states with the same population p: two opposite phases, with suitably unequal weights, realize every $|z| \le \sqrt{p(1-p)}$. Their output entropies are all $e(p)$. Thus

$$f(p) \le F_\Phi(\sigma) \le e(p). \tag{12}$$

The upper bound, rather than equality with f, is what is needed next.

Theorem 6 (Strong convex-roof superadditivity)**.** *For every finite-dimensional channel $\mathcal{N}$ and every joint input state $\rho_{AA'}$,*

$$\widehat{S}_{\Phi\otimes\mathcal{N}}(\rho) \ge \widehat{S}_\Phi(\rho_A) + \widehat{S}_{\mathcal{N}}(\rho_{A'}). \tag{13}$$

The same assertion holds with Φ replaced by any complementary channel Φ^c.

Proof. For a pure $\rho = \psi$, Lemma 5 bounds its output entropy by $e(p)$ plus the average output entropies of the conditional A' branches. By (12), the first term is at least $\widehat{S}_\Phi(\psi_A)$. Those branches decompose $\psi_{A'}$, so their average is at least $\widehat{S}_{\mathcal{N}}(\psi_{A'})$. For mixed ρ, apply this to a minimizing pure decomposition and use convexity of both marginal roofs.

For the complement, let $\mathcal{N}^c$ be a complementary channel to $\mathcal{N}$. For pure ψ, purity of the complete Stinespring output gives

$$S((\Phi^c \otimes \mathcal{N})(\psi)) = S((\Phi \otimes \mathcal{N}^c)(\psi)).$$

Apply Lemma 5 to the right side. Its conditional branches are pure, so $S(\mathcal{N}^c(\psi_j)) = S(\mathcal{N}(\psi_j))$. Thus the same one-step inequality holds for Φ^c. Its pure-output entropy is again $e(p)$, and the preceding argument applies. This use of complement duality is standard [14, Theorem 1, equations (6) and (9)]; see also [15]. □

For a fixed input σ, the maximal ensemble Holevo information is $(S(\mathcal{M}(\sigma)) - \widehat{S}_{\mathcal{M}}(\sigma))/\ln 2$. Subadditivity of the mean output entropy and (13) therefore give its subadditivity at arbitrary joint input marginals, with equality at product input states. This is stronger than the unconstrained maximum in Theorem 1; it uses the established convex-roof framework of [12], not an equivalence asserted without its needed hypotheses.

Corollary 7 (State entanglement cost)**.** *Let $\omega = V\sigma V^*$. For every finite-dimensional bipartite state $\Omega_{BB'EE'}$ with $\Omega_{BE} = \omega$,*

$$E_F(\Omega_{BB':EE'}) \ge E_F(\omega_{B:E}) + E_F(\Omega_{B':E'}).$$

Consequently, for every finite bipartite state τ,

$$E_F(\omega \otimes \tau) = E_F(\omega) + E_F(\tau),$$
$$E_C(\omega) = F_\Phi(\sigma)/\ln 2,$$
$$E_C(\omega \otimes \tau) = F_\Phi(\sigma)/\ln 2 + E_C(\tau).$$

Here E_C is asymptotic LOCC state *entanglement cost in ebits.*

Proof. Positivity and the support of the first marginal imply that Ω is supported on $\operatorname{ran} V \otimes B' \otimes E'$. Lift it through V and use Theorem 6 with $\mathcal{N} = \operatorname{Tr}_{E'}$. The three roofs are exactly the three formation quantities displayed. For product states, product pure decompositions attain the reverse inequality. Iteration gives $E_F(\omega^{\otimes n}) = nE_F(\omega)$, and more generally $E_F(\omega^{\otimes n} \otimes \tau^{\otimes n}) = nE_F(\omega) + E_F(\tau^{\otimes n})$. The state-cost theorem $E_C(\rho) = \lim_n E_F(\rho^{\otimes n})/(n \ln 2)$ [13] proves both cost statements. □

These are states in a Stinespring range, not generally Choi states. The result does not evaluate entanglement cost for simulating the channel. At the GAD boundary the same corollary follows from OpenAI's one-step inequality and the standard correspondence; our block extension supplies the remaining Davies region.

For completeness the required one-use formation functional has a scalar moment description. This is a convenient version of the known axial convex-roof problem [16], not a claim that its geometry is new.

Proposition 8. *With $r(q) = \sqrt{q(1-q)}$ and $\sigma = \left(\begin{smallmatrix} 1-p & z \\ \bar{z} & p \end{smallmatrix}\right)$,*

$$F_\Phi(\sigma) = \min_\mu \left\{ \int e(q)\, d\mu(q) : \int q\, d\mu = p, \quad \int r(q)\, d\mu \geq |z| \right\}, \tag{14}$$

where μ ranges over probability measures on $[0,1]$. In particular $F_\Phi(\operatorname{diag}(1-p,p)) = f(p)$.

Proof. A pure decomposition of σ induces a population measure; the triangle inequality for its coherences gives the moment constraint. Conversely, given such a measure, put $R = \int r\, d\mu$. At every population choose the same average phase factor z/R, with the sign convention matching the upper-right matrix entry. Two opposite phases with probabilities $(1 + |z|/R)/2, (1 - |z|/R)/2$ realize that factor and leave the entropy integral unchanged. If $R = 0$, necessarily $z = 0$ and the endpoint mixture suffices. Compactness gives a minimizing measure. An extreme minimizer can have at most three populations: four supported pieces admit a nonzero signed weight perturbation preserving normalization, the population moment and the radius moment. This also supplies a finite ensemble. For $z = 0$ the constraint is vacuous, reducing the program to the convex envelope. □

Formula (14) is an optimization formula, not an elementary closed form. For example, the sufficient condition $D \geq 0$, $4AD \geq B^2$, with $v(q) = A + Bq + Dq^2$, makes e convex and then $F_\Phi(\sigma) = e(p)$ for every coherence. This is the flat-roof regime already identified by Hellmund and Uhlmann [16, Theorem 2 and Sections IV.C–D]. One quick verification is that $\sqrt{v}$ is a norm of an affine two-vector under this condition, while $g(x^2)$ is convex and nondecreasing [17, 16]. We do not claim this criterion or the flat roof as new. Outside that regime, (14) retains the coherence constraint even though additivity still holds.

5 Classical capacity of the complementary channel

The complementary output is the environment output of a pure Stinespring dilation, including purification of a mixed reservoir when necessary. It is not automatically the experimentally accessible

part of a thermal bath. Its minimal dimension is at most four. To compute its entropy at a diagonal input, choose unit vectors u, v in an environment sector of dimension two, and orthogonal vectors e_+, e_- orthogonal to that sector, with $\langle v, u\rangle = \lambda/\sqrt{(1-a)(1-b)}$. A dilation is

$$V|0\rangle = \sqrt{1-a}|0\rangle|u\rangle + \sqrt{a}|1\rangle|e_+\rangle,$$
$$V|1\rangle = \sqrt{b}|0\rangle|e_-\rangle + \sqrt{1-b}|1\rangle|v\rangle.$$

The positivity constraint makes the prescribed inner product feasible. If its denominator vanishes then $\lambda = 0$ and the irrelevant overlap may be chosen arbitrarily. Tracing the environment gives (1). For input $\mathrm{diag}(1-p, p)$ the environment spectrum is

$$a(1-p), \quad bp, \quad \frac{w(p) + \sqrt{w(p)^2 - 4d(p)}}{2}, \quad \frac{w(p) - \sqrt{w(p)^2 - 4d(p)}}{2}, \tag{15}$$

where

$$w(p) = (1-a)(1-p) + (1-b)p, \qquad d(p) = p(1-p)((1-a)(1-b) - \lambda^2).$$

Indeed the remaining block is the mixture of $|u\rangle\langle u|$ and $|v\rangle\langle v|$ with the indicated trace and determinant. Let $H_E(p)$ be the entropy in nats of the four probabilities in (15).

Theorem 9. *For every qubit Davies channel in (2),*

$$C(\Phi^c) = \chi(\Phi^c) = \frac{1}{\ln 2} \max_{0 \le p \le 1} \{H_E(p) - f(p)\}.$$

At most two opposite-phase population pairs attain the one-use maximum. For every finite-dimensional $\mathcal{N}$,

$$\chi(\Phi^c \otimes \mathcal{N}) = \chi(\Phi^c) + \chi(\mathcal{N}),$$
$$C(\Phi^c \otimes \mathcal{N}) = \chi(\Phi^c) + C(\mathcal{N}),$$
$$S_{\min}(\Phi^c \otimes \mathcal{N}) = \min_p e(p) + S_{\min}(\mathcal{N}).$$

Proof. For $U_\theta = \mathrm{diag}(1, e^{i\theta})$, let W_θ act as the identity on u, v and with phases $e^{-i\theta}, e^{i\theta}$ on e_+, e_-. Then $VU_\theta = (U_\theta \otimes W_\theta)V$. Hence Φ^c is phase covariant, and averaging the input phase cannot decrease its output entropy. At mean population p that entropy is therefore at most $H_E(p)$, while the mean pure-output entropy is at least $f(p)$. The same opposite-phase ensemble used in Theorem 1 attains both bounds.

The proof of Theorem 6 established the full one-step inequality for Φ^c. Repeating the branch-ensemble proof of Theorem 1, with H_E replacing $h \circ t$, gives the arbitrary-partner Holevo and minimum-entropy identities. Regularization gives the capacity identity. □

The function H_E is concave, being entropy along an affine family of states; therefore $H_E - f$ is concave. A sampled lower hull f_N gives the same one-sided discretization error as Proposition 14, since only the error in f enters that argument. The limits provide useful checks: an identity channel has a constant complement of capacity zero, whereas a constant qubit-output channel has complementary capacity one bit.

6 Parallel simulation of the complementary channel

This section concerns the resources needed to simulate $\mathcal{N} = \Phi^c$, with its full Stinespring output. Let

$$G_{\mathcal{M}}(\sigma) = E_F((\mathrm{id}_R \otimes \mathcal{M})(\psi_\sigma)), \qquad E_F^{\mathrm{ch}}(\mathcal{M}) = \max_\sigma G_{\mathcal{M}}(\sigma),$$

where ψ_σ is a purification and E_F uses nats. Reference isometries do not affect this definition. The parallel simulation cost E_C^{par} is the asymptotic number of shared ebits per use required by LOCC protocols whose simulated tensor-power channel converges in diamond norm; classical communication is free. The theorem of Berta et al. gives

$$E_C^{\mathrm{par}}(\mathcal{M}) = \lim_{n\to\infty} \frac{E_F^{\mathrm{ch}}(\mathcal{M}^{\otimes n})}{n \ln 2}. \tag{16}$$

This operational theorem is external [19, Theorem 12]. Sequential or adaptive tests constitute a different task [20].

We recall two standard formation facts. The Kraus characterization is

$$G_{\mathcal{M}}(\sigma) = \inf_{\{L_j\}} \sum_j \mathrm{Tr}(L_j \sigma L_j^*) S\left(\frac{L_j \sigma L_j^*}{\mathrm{Tr}(L_j \sigma L_j^*)}\right),$$

with zero branches omitted. Pure-state ensemble decompositions of the purified output correspond to Kraus representations; one first argues on the support of σ and extends the representation or takes continuous limits at singular inputs. For every fixed representation the entropy perspective is concave in σ. Its infimum is concave as well: bound each component of a mixture by its own infimum before taking the infimum over the common representation. Thus $G_{\mathcal{M}}$ is concave. It is continuous by continuity of finite-dimensional E_F and of the canonical purification constructed from $\sqrt{\sigma}$, so a maximum exists. Covariance and local-unitary invariance then imply that G_{Φ^c} has a diagonal maximizer, by averaging the input phase.

The other fact is subadditivity of channel formation [2, Appendix C]:

$$E_F^{\mathrm{ch}}(\mathcal{M} \otimes \mathcal{L}) \le E_F^{\mathrm{ch}}(\mathcal{M}) + E_F^{\mathrm{ch}}(\mathcal{L}). \tag{17}$$

For completeness, choose Kraus decompositions optimal, or arbitrarily close to optimal, for the two marginal input states of a joint purified input. Their product yields an admissible pure decomposition of the joint output. For its classical Kraus labels I, J and quantum outputs B, B', entropy subadditivity and conditioning give

$$S(BB'|IJ) \le S(B|IJ) + S(B'|IJ) \le S(B|I) + S(B'|J).$$

Completeness of the other instrument makes each final term the corresponding local Kraus objective. Maximizing over inputs proves (17).

Theorem 10 (Complementary simulation)**.** *For every Davies channel* Φ *in* (2) *and every finite channel* $\mathcal{M}$*,*

$$\begin{aligned} E_F^{\mathrm{ch}}(\Phi^c \otimes \mathcal{M}) &= E_F^{\mathrm{ch}}(\Phi^c) + E_F^{\mathrm{ch}}(\mathcal{M}), \\ E_C^{\mathrm{par}}(\Phi^c) &= E_F^{\mathrm{ch}}(\Phi^c)/\ln 2, \\ E_C^{\mathrm{par}}(\Phi^c \otimes \mathcal{M}) &= E_F^{\mathrm{ch}}(\Phi^c)/\ln 2 + E_C^{\mathrm{par}}(\mathcal{M}). \end{aligned}$$

An exact scalar formula is

$$E_C^{\mathrm{par}}(\Phi^c) = \frac{1}{\ln 2} \max_{0\le p\le 1} f_p(t(p)), \tag{18}$$

where f_p is the lower convex envelope of the pure-output entropy for the reversed Davies parameters

$$a_p = \frac{pb}{1-t}, \qquad b_p = \frac{(1-p)a}{t}, \qquad k_p = \frac{kp(1-p)}{t(1-t)}, \qquad \lambda_p^2 = \frac{\lambda^2 p(1-p)}{t(1-t)}, \quad t = t(p). \tag{19}$$

Endpoint and constant pure-output exceptions are specified in the proof.

Proof. For $0 < p < 1$ and $0 < t < 1$, apply the dilation of Section 5 to half of $\sqrt{1-p}|00\rangle + \sqrt{p}|11\rangle$. Grouping the resulting pure vector by its B output gives an isometry W_p from a qubit to RE:

$$W_p|0\rangle = \frac{\sqrt{(1-p)(1-a)}|0\rangle|u\rangle + \sqrt{pb}|1\rangle|e_-\rangle}{\sqrt{1-t}},$$
$$W_p|1\rangle = \frac{\sqrt{(1-p)a}|0\rangle|e_+\rangle + \sqrt{p(1-b)}|1\rangle|v\rangle}{\sqrt{t}}.$$

Its columns are orthonormal. Tracing E gives a qubit channel Ψ_p with parameters (19); in particular $1 - a_p - b_p = k_p$ and $0 \le \lambda_p^2 \le k_p$. This is the key closure property. The reference–environment output state is exactly

$$\omega_p = (\mathrm{id}_R \otimes \Phi^c)(\psi_p) = W_p \,\mathrm{diag}(1-t, t) W_p^*.$$

Corollary 7 therefore gives strong formation additivity for ω_p with every finite bipartite partner, and $G_{\Phi^c}(\mathrm{diag}(1-p, p)) = f_p(t)$.

Choose a diagonal input maximizing G_{Φ^c} and an optimizing purified input for $\mathcal{M}$, with output state τ. Product inputs, together with the established state additivity, give

$$E_F^{\mathrm{ch}}(\Phi^c \otimes \mathcal{M}) \ge E_F(\omega_p \otimes \tau) = E_F^{\mathrm{ch}}(\Phi^c) + E_F^{\mathrm{ch}}(\mathcal{M}).$$

Equation (17) proves equality. Iteration and (16) prove both operational identities and (18).

At $p = 0, 1$ the reference is pure and $G = 0$; a zero maximizing formation value also gives the product lower bound by local discard. If $t = 0$ or 1 at an interior p, the original channel is constant with pure output. Its complement is isometric, $G(\mathrm{diag}(1-p, p)) = h(p)$ and its cost is one ebit. Pure maximal entanglement gives its strong product lower bound. These cases define the missing values of (18); all other boundary cases follow from the displayed construction or continuity. □

This proof uses an optimizing reference–output state in a different Stinespring range; complement duality alone does not transfer simulation cost. The construction is the standard normalized reverse-channel (weighted-transpose or Petz) construction [21], with its explicit Davies parameters verified here. At the GAD boundary the same deduction uses the antecedent one-step inequality [3]. The added dephasing and nonunital parameters are the extension.

Corollary 11 (An elementary rate)**.** *Put $x = \lambda^2$ and*

$$c(a, b) = k + 2ab - 2\sqrt{ab(k + ab)}.$$

If $c(a, b) \le x \le k$, then

$$E_C^{\mathrm{par}}(\Phi^c) = h\left(\frac{1+\lambda}{2}\right) / \ln 2.$$

A Bell input maximizes channel formation in this region.

Proof. The threshold c is the known Hellmund–Uhlmann flat-concurrence threshold, a sufficient condition for the entropy roof used here [16, Equation (29) and Section V.B]. With $L = p(1-p)/(t(1-t))$, the reverse parameters satisfy $k_p = Lk$, $a_p b_p = Lab$, $\lambda_p^2 = Lx$, and hence $c(a_p, b_p) = Lc(a, b)$. Their entropy roof is therefore flat in the stated region. Direct substitution yields

$$f_p(t) = e_p(t) = g((1-x)p(1-p)).$$

The maximum is at $p = 1/2$, proving the formula. The exceptional constant pure-output case has the same value one; continuity covers the remaining endpoints. □

The threshold itself is an antecedent, not a new convex-roof criterion or a necessary boundary for entropy flatness or for this simulation formula. For example, $a = 0.1, b = 0.3, \lambda^2 = 0.5$ has complementary parallel cost about 0.600876037 ebits per use. This determines the complementary simulation rate, not the simulation cost of Φ.

There is also a sufficient low-coherence regime with a simpler optimization. Define the classical conditional entropy

$$J(p) = h(p) + (1-p)h(a) + ph(b) - h(a+kp).$$

The endpoint mixture in every reversed roof gives $f_p(t) \le J(p)$, by the classical entropy chain rule. If p_0 maximizes J, is interior, and has $0 < t_0 = t(p_0) < 1$, then

$$\lambda^2 \le \frac{k^2 p_0(1-p_0)}{t_0(1-t_0)} \quad \Longrightarrow \quad E_C^{\mathrm{par}}(\Phi^c) = J(p_0)/\ln 2. \tag{20}$$

Indeed, the quadratic coefficient of the reversed determinant at p_0 is $L(\lambda^2 - Lk^2) \le 0$. Since g is increasing and concave, e_{p_0} is concave. Its convex envelope is its endpoint chord, attaining the upper bound $J(p_0)$. Equation (20) is sufficient; no complete transition diagram is asserted.

Proposition 12 (Strict separation from Choi-state cost)**.** *For $a = 1/10, b = 3/10, \lambda^2 = 1/10$, let $\mathcal{N} = \Phi^c$ and let $J_{\mathcal{N}}$ be its normalized Choi state. Then*

$$E_C^{\mathrm{par}}(\mathcal{N}) > E_C(J_{\mathcal{N}}).$$

In particular a Bell input does not maximize $E_F^{\mathrm{ch}}(\mathcal{N})$.

Proof. At $p = 1/2$ the reversed parameters have $a_p = 1/4$, $b_p = 1/8$, $k_p = 5/8$ and $\lambda_p^2 = 5/48$. The quadratic coefficient is $5/48 - 25/64 = -55/192 < 0$, so it remains negative in an open interval of p. There the endpoint-chord argument gives $G_{\mathcal{N}}(p) = J(p)$ exactly. But

$$J'(1/2) = h(3/10) - h(1/10) - \tfrac{3}{5}\ln(3/2) = \tfrac{1}{10}\ln\left(\frac{2^6 3^9}{7^7}\right) > 0,$$

because $1259712 > 823543$. Thus an input with $p > 1/2$ has strictly larger formation than the Choi input. Theorem 10 identifies the channel cost with its maximum formation, whereas Corollary 7, applied to $W_{1/2}$, identifies the Choi-state cost with its formation. □

Consequently this channel cannot be simulated exactly by LOCC using one copy of its own Choi state per call: preparing many such states at their state cost would contradict the strict inequality. Other resource states are not excluded. The example has $\lambda^2 > ab$ and is nonunital, so it does not merely restate the covariant epolarizing example of [20]. Approximate sampled values are 0.704192652 ebits at the Bell input and 0.704712865 near $p = 0.516949108$; the strict proof does not use these values.

7 Finite-block consequences

The preceding result also applies to nonidentical channels in (2). Write e_i, f_i, t_i for the corresponding functions of channel Φ_i.

Corollary 13. *For every state ρ on n qubits, with marginal excitation probabilities p_i,*

$$S((\Phi_1 \otimes \cdots \otimes \Phi_n)(\rho)) \geq \sum_{i=1}^{n} f_i(p_i).$$

Proof. For a pure input split at the first site and use (9). Apply induction on each conditional remaining state and combine its populations using convexity of each f_i. At the first site $e_1 \geq f_1$. A pure-state decomposition, entropy concavity and convexity of the f_i give the mixed-input statement. □

This yields an exact accounting of the Holevo deficit. Let a pure ensemble have output states ω_ℓ, mean $\bar{\omega}$, populations $p_{i\ell}$ and means $\bar{p}_i$. Put

$$c_i = (\ln 2)\chi(\Phi_i), \quad \sigma = \bigotimes_i \operatorname{diag} \bar{\omega}_i, \quad R_\ell = S(\omega_\ell) - \sum_i f_i(p_{i\ell}) \geq 0.$$

Using quantum relative entropy in nats, direct subtraction gives

$$\begin{aligned} \sum_i c_i - (\ln 2)\chi(\text{ensemble}) = &\sum_i [c_i - h(t_i(\bar{p}_i)) + f_i(\bar{p}_i)] \\ &+ D(\bar{\omega}\|\sigma) + \sum_\ell w_\ell R_\ell \\ &+ \sum_i \left[\sum_\ell w_\ell f_i(p_{i\ell}) - f_i(\bar{p}_i) \right]. \end{aligned} \tag{21}$$

Every term is nonnegative. In particular, $D(\bar{\omega}\|\sigma) = \sum_i h(t_i(\bar{p}_i)) - S(\bar{\omega})$. The support convention is harmless: a zero marginal basis probability gives zero support of $\bar{\omega}$ on that sector. Equation (21) is a consequence of the population bound and standard relative-entropy identities. It extends the usual finite-block deficit method, including that used for amplitude damping in [2]; it does not imply uniqueness of optimizers or a uniform positive gap.

8 Scalar computation and examples

The envelope in (10) can be evaluated by the lower convex hull of the sampled curve. The next bound makes the discretization error explicit, separately from floating-point error.

Proposition 14. *Suppose e is L-Lipschitz on $[0,1]$. For a uniform grid of mesh $1/N$, let f_N be the lower convex envelope of its sampled points, linearly extended between hull vertices. If*

$$C_N = \frac{1}{\ln 2} \max_p \{h(a+kp) - f_N(p)\},$$

then

$$C_N \leq C(\Phi) \leq C_N + \frac{L}{2N \ln 2}.$$

Proof. The piecewise-linear interpolant $\widetilde{e}_N$ satisfies $|\widetilde{e}_N - e| \le L/(2N)$: at relative position θ in a cell the error is at most $2L\theta(1-\theta)/N$. Taking lower convex envelopes preserves uniform error bounds. Moreover $f_N \ge f$, because mixtures of sampled populations form a subset of all mixtures; and $\operatorname{co}\widetilde{e}_N = f_N$. Thus $0 \le f_N - f \le L/(2N)$, which gives the claim. □

For parameters with $m = \min_p v(p) > 0$, write

$$v(p) = A + Bp + Dp^2, \quad A = a(1-a), \quad B = k(1-2a) - \lambda^2, \quad D = \lambda^2 - k^2.$$

The minimum is found from the endpoints and any interior vertex of this quadratic. A valid Lipschitz constant is

$$L = \max\{|B|, |B+2D|\}\, g'(m), \qquad g'(m) = \frac{2\operatorname{atanh}\sqrt{1-4m}}{\sqrt{1-4m}},$$

with limit 2 at $m = 1/4$. The function g' is nonincreasing; this follows by differentiating $\operatorname{atanh} x/x$. When $m = 0$ one can instead use uniform continuity; the displayed finite-L bound is not asserted at that boundary.

For $k > 0$, on a hull segment of slope s, the maximizing population is the projection onto that segment of

$$p = \frac{(1+\exp(s/k))^{-1} - a}{k}.$$

This follows by setting $k\ln((1-t)/t) = s$ and using concavity. The case $k = 0$ has zero capacity.

A strict interior example is

$$a = \frac{1}{10}, \quad b = \frac{3}{10}, \quad k = \frac{3}{5}, \quad \lambda^2 = \frac{1}{2}.$$

It is nonunital, has additional dephasing, and is not entanglement breaking. In fact the Choi partial transpose is positive exactly when $\lambda^2 \le ab$, which fails here. Its Choi rank is four, and $\min v = 5/56 > 0$, so it admits no pure output. The example is therefore not covered merely by the unital, entanglement-breaking, pure-output or $\lambda^2 = k$ boundary cases.

a	b	λ^2	capacity (bits, approximate)	mean population
0.10	0.30	0.60	0.503934607	0.468332
0.10	0.30	0.50	0.407849860	0.470350
0.10	0.30	0.10	0.296671803	0.471876
0.02	0.30	0.514624	0.436551160	0.441065

These are reproducible floating-point illustrations on a grid with $N = 200000$, not machine-certified intervals. Proposition 14's exact-arithmetic error bounds are below 3.8×10^{-6} bits for these cases, before rounding error. The last example's sampled hull optimizer mixes population 0 with approximately 0.958865, illustrating why a single population pair need not suffice. No exact three-signal necessity or global transition boundary is inferred from that numerical observation. For the same four channels, a separate computation with $N = 60000$ gives complementary capacities approximately 0.492274059, 0.562095952, 0.906804474 and 0.469017072 bits. Direct Stinespring matrices reproduce (15) and the phase covariance in 500 random cases, with maximum entry or eigenvalue error below 5.0×10^{-16}. These are floating-point illustrations, not machine-certified intervals. For $a = 0.02, b = 0.30, \lambda^2 = 0.514624$, a sampled version of (14) at $p = 0.44$ gives state formation costs about 0.466960 ebits at $z = 0$, 0.467338 at $z = 0.25$, and 0.469937 at $z = 0.45$. The coherence constraint matters; the scalar roof $f(p)$ alone does not describe all states. The linear program uses 20001 population samples and reconstructs a feasible phase ensemble up to numerical tolerance; no continuous optimization certificate is claimed.

9 Verification, dependence and scope

The proof uses Lemma 2 as an external result and gives all subsequent matrix and channel arguments explicitly. Some minimum-output-entropy subregions are also covered by Fukuda's composition criterion [18]; that criterion is not a generic closure theorem for unrestricted Holevo additivity under composition. A root-agent exploratory screen tested 16000 random rectangular block instances without a violation. Separately, an Astra candidate audit tested 9000 block instances, their 27000 vertex instances, 1200 channel outputs with arbitrary finite partner samples, and 300 ensembles. Its smallest residual was of order 10^{-14} in equality cases. These floating-point tests are error-finding aids, not proofs.

A tempting alternative argument—that extra dephasing must increase the block entropy by at least the scalar entropy increase—failed a numerical check and is not used. The valid argument preserves the scalar entropy across the three vertices and invokes concavity only after their separate estimates.

The present family excludes the permitted phase-covariant region $\lambda^2 > 1-a-b$ and channels with negative longitudinal multiplier. We determine the state costs in Corollary 7 and the complementary parallel simulation cost in Theorem 10. We do not determine the original channel's simulation cost, adaptive simulation cost, quantum or private capacity, interacting multiqubit Davies dynamics, or finite-error coding rates. Memoryless use is essential to the operational interpretation. No claim of exhaustive literature coverage or worldwide priority accompanies the stated extension; its mathematical scope is explicit and its immediate antecedents are identified.

AI assistance and review. The author directed an AI-assisted research process. Codex developed the extension, proofs, exposition and scalar calculations. A separately dispatched reviewer requested as GPT-6 Astra with xhigh effort audited frozen proof dossiers and an earlier capacity-only manuscript. The reviewer has prior involvement and is not blind to the work. A subsequent integrated version was also reviewed. The present version further adds complementary parallel simulation, reverse-channel closure and the strict Choi-cost separation; every manuscript assessment must identify its exact input version, and an earlier assessment does not automatically apply to these additions. Model runtime identity was not independently attested. No human peer review or predetermined numerical grade is claimed.

Availability. The supplement contains the proof dossier, numerical scripts and reports, and source-retrieval metadata. Original implementation code is released under CC0; the manuscript is offered under CC BY 4.0. External papers retain their authorship and licences. The scalar scripts do not implement directed-rounding interval certification.

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